

Boundary-Trace Fixation under Admissibility

A Conditional Theorem for Environmental Role Anchoring

Amos Jay Maley

May 2026; scientific revision October 2026

Abstract

Boundary-trace fixation concerns a singular constructional role not fixed by the complete licensed non-boundary input. We prove an environmental fixation theorem on the original token carrier. Canonical complete qualification is derived from the frozen role, and an exact realization criterion determines when channel data can represent it. The channel-to-role identification, joint dependencies, and lawful transport must be established from the original conditions. When the resulting role-qualified trace has exactly one positive token, that token is the unique admissible occupant, with the other role assignments held fixed. The trace records the original qualification; declaring a predicate or restricting a symmetry group does not confer it. Separately, a sound licensed trace, when singleton-positive on the relevant token or class carrier, supplies an admissible selected token or a unique trace-fixed role class without having to classify every originally admissible alternative. An originally required trace discharges the stronger completeness bridge from its necessity and positive license.

The converse distinguishes failure of a specified trace operator from impossibility for every licensed rule. We prove an exact stabilizer criterion for structural selection, a fibre criterion for environmental determination, and a descent-and-lift criterion separating token fixation from role-class fixation. These results retain lawful joint and richer inputs. In the regime where the complete internal-plus-indexical input cannot warrant fixation, any successful remaining same-domain route requires independently licensed boundary data or boundary grounds. Dependence of the answer and dependence of its warrant are distinguished. Neither can acquire its authority from the downstream confirmation whose role it is meant to fix. Later discovery of an original qualification is distinguished from later alteration of the original act. The theorem establishes neither a physical channel's existence nor a dynamical selection mechanism.

Contents

1	Original question and proof order	2
2	Internal non-selection and the actual selector input	4
3	Joint source exhaustion and environmental dependence	5
4	The environmental trace and its original-role authority	7
5	The partial token operator and the class comparison	10

6	Boundary-Trace Fixation Theorem	13
7	Converse, completeness, and lawful enrichment	15
8	Class III and exact downstream reception	18
9	A structural witness and its limits	19
10	Objections and the exact receiving burden	20
11	Scope and challenge conditions	22
12	Conclusion	22
A	Source bindings and receiving ledger	23
B	Scientific scope of this revision	24
	References	26

1 Original question and proof order

The question is which already available token occupies one specified constructional role. It is prior in the order of justification to confirmation using that role. The question is not settled merely because a token is physically present, its inscription is intelligible, or a theory predicts its content. Nor is every role question a token question: some ask only for a class of equivalent role assignments. This paper retains the singular token target of the May 2026 manuscript and states separately what can be exported to a class-valued receiver.

Kernel governance is necessary for non-degenerate determinate construction, including unobserved physical incidence. Reference retains the bearer and original conditions; standing concerns actual qualification for the specified use; admissibility concerns their joint satisfaction and warranted reliance; irreversibility retains the original occurrence against substitution. These functions belong to the original realization, and descriptions answer to them. They are not made optional by choosing a framework or an observer. The Kernel's constitutive and reconstruction arguments establish that necessity [2]. The application here must additionally establish its substantive channel and role premises. Necessary governance alone does not supply a particular channel, a unique role occupant, or an empirical law.

Definition 1.1 (Target fixed domain). Retain the target tuple

$$\mathcal{D}_{\text{env}} = (\mathcal{C}, \mathcal{L}, \mathcal{S}, \text{Rep}, \text{Out}, \text{Adm}, \text{Aut}, \mathcal{E}, \text{Ch}).$$

Here $\mathcal{C}$ is the construction system, $\mathcal{L}$ its role scheme, $\mathcal{S}$ its constructed structures, and Rep, Out the original report and outcome-profile interfaces. The environment $\mathcal{E}$, channel package Ch , admissibility relation Adm , and lawful comparisons Aut belong to the original domain. For the role ℓ , fix a set

$\mathcal{S}_\ell \subseteq \mathcal{S}$ of candidate token identities and a background partial assignment R on the other roles. Write

$$R_s = R[\ell \mapsto s] \quad (s \in \mathcal{S}_\ell).$$

The background includes the relevant occurrence, conditions, original consumers, and joint compatibility requirements. It is held fixed whenever uniqueness of the occupant of ℓ is claimed. A token identity is not identified with a coordinate string used to present it.

Definition 1.2 (Complete original role requirements). Let d denote the complete original input for this role problem. Its independently specified requirement family U_ℓ includes every condition required for the claimed role use: candidate eligibility, channel incidence, compatibility with the fixed other assignments, and any report, continuation, transport, historical, or application-domain condition required by that use. Let $A_u(d, s)$ express satisfaction of requirement u , and put

$$J_\ell(d, s) \iff \bigwedge_{u \in U_\ell} A_u(d, s), \quad F_\ell(d) = \{s \in \mathcal{S}_\ell : J_\ell(d, s)\}.$$

The domain meanings of the A_u , their coverage of the original task, and their joint evaluation at this same input are supplied before testing a proposed trace. Thus $J_\ell(d, s)$ expresses $\text{Adm}(R_s)$ for this original role use. No singleton condition is assumed in this definition. A proposal cannot define these requirements by its desired output or omit an original consumer to obtain uniqueness.

The conjunction records original conditions obtaining together. It is not a rule inferring joint admission from separate component certificates. A condition involving two records, for example, must be evaluated on those records together; qualification of each record alone does not settle their compatibility. The family can be infinite and need not be decidable. A concrete application must prove its coverage from the original task, rather than from the number of entries in a ledger.

Remark 1.3 (Original qualification, declaration, and knowledge). Declaration fixes the question being asked and the data a proposed rule claims to use. It does not create a physical incidence. An originally qualifying but unread channel can later be discovered or certified. Adding or changing a channel that the original act lacked instead supplies a different input or occurrence. Neither case changes the original qualification. If the original question also asks what was warranted to an investigator at the time of use, that epistemic requirement is another original condition and must be assessed as such.

Definition 1.4 (Import discipline). An imported result is used only after identifying its original bearer, complete question, candidate and result types, licensed data, lawful transformations, and joint dependencies with the receiver. Qualification for a specified use is bivalent; complete role content need not be one bit. Faithful representation preserves every original required answer and its semantic domain. A result about one role, quotient, or occurrence does not automatically authorize a different one. Later evidence and fresh licensed constructions are permitted; retroactive substitution of their qualification for a failed original incidence is not.

The proof order is original incidence, then faithful trace representation, then singleton fixation and lawful transport. Exhaustion concerns every rule over the complete licensed input, not a finite list of familiar methods. Appendix A separates historical imports from the explicitly versioned current receiving results. The mathematical arguments use classical logic on set-sized carriers; Boolean qualification and semantic existence do not assert decidability, an executable selector, or a physically realized channel.

2 Internal non-selection and the actual selector input

Let $|\mathcal{S}_\ell| \geq 2$ before role-fixing constraints are applied. The internal reduct of the construction system will be denoted $\mathcal{F}$; it is not an additional source of data.

Definition 2.1 (Role assignment). A role assignment is a partial map $R : \mathcal{L} \rightharpoonup \mathcal{S}$. Comparisons at ℓ vary R_s over the original candidate fibre while holding the other assigned roles fixed. Uniqueness at ℓ does not assert uniqueness of the whole role scheme.

Definition 2.2 (Reports, outcomes, and neutrality). The interfaces $\text{Rep}(R_s)$ and $\text{Out}(R_s)$ give the complete report content and outcome profiles required by their original contexts. Outcome-completeness means that the relevant interface is specified for each candidate on its original domain of use. Outcome-profile neutrality is the separate assertion

$$\text{Out}(R_s) = \text{Out}(R_t)$$

for all compared candidates and every original admissible outcome-ordering. Report neutrality is the corresponding equality for Rep . Equality of these interfaces does not establish equality of all original role consumers unless a coverage argument shows that they exhaust those consumers.

Definition 2.3 (Internal non-selectivity). The original role problem is internally non-selective if no rule warranted solely by its licensed internal input fixes a unique occupant while preserving the original role requirements and interfaces. A stronger exhausted regime, used below, excludes such a rule on the *joint* internal-plus-indexical/provenance input. Failure of each component to select by itself is not this joint hypothesis.

Definition 2.4 (Lawful presentation action). Let a group G act on a family Z of licensed input presentations and on the candidate output identities S . The family and actions are identified from the original data and their lawful transport. A candidate family $F(z) \subseteq S$ is covariant if $F(gz) = gF(z)$. A partial selector is a map $f : D \rightarrow S$, where $D \subseteq Z$ is invariant, with

$$f(z) \in F(z), \quad f(gz) = gf(z).$$

Its actual input is z , not an already chosen candidate. Its domain is part of the covariance claim. Write $G_z = \{g : gz = z\}$ for the input stabilizer and $\text{Fix}_S(G_z)$ for the candidates it fixes.

An original incidence may be determinate even when an available-data interface forgets its discriminator. A symmetry of that coarse interface must not be presented as a symmetry of the complete occurrence. Conversely, parameters that are actually part of the input must transform with it. One cannot keep a preferred token name fixed while calling a permutation that moves it a symmetry of that complete input.

Definition 2.5 (Symmetry saturation). At an input z , an eligible set is symmetry-saturated for a no-selection argument if every eligible candidate is moved by some element of G_z ; equivalently,

$$F(z) \cap \text{Fix}_S(G_z) = \emptyset.$$

Transitivity on an eligible set with at least two elements is sufficient. Merely having two trace-positive candidates is not a symmetry-saturation condition.

Lemma 2.6 (Internal neutrality blocks an unanchored selector). *No eligible equivariant selector is defined at an input whose eligible set is symmetry-saturated. In particular, a transitive internal candidate problem with at least two candidates has no internal singular selector from that input alone.*

Proof. If $f(z) = s$, then for each $g \in G_z$, equivariance gives $gs = gf(z) = f(gz) = f(z) = s$. This places s in the fixed-point intersection, contrary to its emptiness. The transitive case has no fixed candidate because another candidate is reachable by an input-preserving transformation. $\square$

Theorem 2.7 (Exact uniqueness criterion for structural selection). *For an invariant D and covariant F , set*

$$K_z = F(z) \cap \text{Fix}_S(G_z).$$

There is exactly one eligible equivariant selector on D if and only if K_z is a singleton for every $z \in D$.

Proof. Stabilizers conjugate under transport, giving $K_{gz} = gK_z$. If each set is singleton, assign its unique member. This defines an eligible function without choosing among alternatives, and the displayed transport law proves equivariance and uniqueness.

Conversely, let f be the unique selector. Its value belongs to K_z by the preceding lemma's calculation. Given $t \in K_z$, define another selector on the orbit of z by $f_t(gz) = gt$, and leave f unchanged elsewhere. If $gz = hz$, then $h^{-1}g \in G_z$, so $gt = ht$; the definition is well-defined. Covariance supplies eligibility, and transport supplies equivariance. The orbit and its complement are invariant. Uniqueness of f now gives $t = f_t(z) = f(z)$. Thus K_z is singleton. The empty-domain case has the unique empty function and satisfies the pointwise condition vacuously. $\square$

This is the selector criterion received from AMetric Theorem 5.4 [3], with its input and eligibility hypotheses explicit. It proves structural uniqueness. Identifying the selected value with the original physical or constructional role still requires that role's incidence warrant. Conversely, existence of a set-theoretic function need not give an executable or physically available procedure.

Remark 2.8 (Outcome comparison is not an anchor by itself). Neutral reports or outcomes do not select a candidate. More generally, downstream confirmation cannot supply the first warrant for the very role used to interpret that confirmation. This does not exclude detection events, evidence about an original channel, or original physical outcome data. Section 4 states the relevant independence requirement.

3 Joint source exhaustion and environmental dependence

Definition 3.1 (Licensed sources and their support). Resolve the complete original input into internal data i , indexical/provenance data a , and boundary/channel data c . The realized family Z is a subset of these joint tuples; arbitrary recombination is not assumed. Every source used by a same-domain rule must be accounted for in this original family, including its interpretation and warrant. A relation depending on several coordinates is joint data, not an extra independent primitive by default. Let $p : Z \rightarrow X = p(Z)$ forget c and retain the complete joint (i, a) input.

This support description is a coverage obligation for the actual target tuple, not a physical taxonomy proving that every conceivable domain contains only three kinds of object. If relevant original data were omitted, the claimed coverage fails. If genuinely new data or a new question are supplied, their change is stated explicitly. A derived predicate of data already licensed in the original joint input is not automatically a domain reset.

Lemma 3.2 (Source normal form by exact factorization). *Let $f : Z \rightarrow S \sqcup \{\perp\}$ be the totalized answer of an originally warranted partial role rule, where $\perp$ records its semantic domain of unavailability. There exists a unique $\bar{f} : X \rightarrow S \sqcup \{\perp\}$ with $f = \bar{f} \circ p$ if and only if*

$$p(z) = p(z') \implies f(z) = f(z').$$

If no such factorization exists, some pair with the same complete non-boundary input requires different answers or different semantic availability. The difference is carried by the boundary part of the joint input. This assertion includes combinations of sources.

Proof. A factored function has equal values on each fibre. Conversely, assign to $x \in p(Z)$ the common answer of its preimages. The condition makes it well-defined, and the attained-image restriction gives existence and uniqueness. Failure of the condition supplies the stated pair. Since the complete tuple has no other omitted coordinate, that pair differs in its boundary data or their joint incidence. No step selects a representative of a fibre or assumes a finite enumeration of rules. $\square$

Theorem 3.3 (Asymmetry exhaustion for the original role). *With complete source coverage established, every warranted role-fixing rule either factors through the joint internal-plus-indexical input or essentially uses licensed boundary information. A claimed use of a further source must establish its original inclusion and warrant, or declare the changed input/question. An unsupported source does not acquire authority through its name or output.*

Proof. Apply the factorization alternative to the actual role answer, including definedness. In the first case the original non-boundary input determines that answer. In the second, the pair witness proves that a rule on that input alone cannot reproduce it. The accounted-for remaining data are boundary data. Information determination does not itself establish warrant; the original incidence and rule soundness are required in both cases. A proposal outside the covered input has either identified a missing original condition, changed the task, or left a warrant unestablished. These cases do not constitute a new warranted answer to the unchanged complete question. $\square$

Corollary 3.4 (Conditional forcing of the environmental route). *If the complete joint internal-plus-indexical/provenance input supplies no warranted singular answer for the original role, every remaining warranted same-domain fixation requires boundary data or boundary grounds for its original-role warrant. If its value or semantic availability fails to descend, Lemma 3.2 supplies an information-loss witness. If it descends, the boundary may instead be needed to warrant the descended answer. Neither possibility proves that a qualifying boundary source exists.*

Proof. Suppose a warranted full-input fixation used neither boundary information nor boundary grounds. Its answer and its independent original-role justification would then be available on the complete joint non-boundary input, contrary to the hypothesis. Thus at least one requires the boundary. When the answer does not descend, the factorization lemma gives the stated pair witness. When it does descend, that semantic factorization alone does not establish a reduced-input warrant; the missing ground can still be boundary incidence. This argument does not infer nonfactorization merely from absence of a reduced-input license. $\square$

Remark 3.5 (Joint information is not marginal selection). Two non-singleton constraints can have a singleton intersection: on $\{a, b, c\}$, the constraints $\{a, b\}$ and $\{b, c\}$ jointly retain only b . This does not supply an admissibility warrant, but it shows why exhaustion must test the entire licensed relation. A channel may likewise fix a role only together with original context or provenance data.

The loci are a cover of support, not mutually exclusive categories. Class III denotes a case in which boundary determination or boundary warrant is necessary after the whole non-boundary reduct has been tested; it need not be independent of every other coordinate.

4 The environmental trace and its original-role authority

Definition 4.1 (Environment and joint channel incidence). An environment is an interaction partner not exhaustively represented by the internal reduct but included in the original environmental question. A channel package specifies the actual relation between candidate tokens, that environment, the original occurrence, and every relevant protocol or compatibility condition. Write $C_d(s)$ for this joint channel incidence. The trace computation and its license must follow from that incidence and independently justified domain rules. Neither a channel name nor separate admission of its components supplies the joint relation.

Definition 4.2 (Typed trace and exact role reception). A typed boundary trace is a bivalent predicate

$$\text{Tr}_d : \mathcal{S}_\ell \rightarrow \{0, 1\}$$

representing a declared channel condition derived from C_d . For it to be *role-exact* at the original input, establish

$$\text{Tr}_d(s) = 1 \iff J_\ell(d, s) \quad (s \in \mathcal{S}_\ell). \quad (4.1)$$

The two directions are separate receiving obligations. Soundness says that a positive trace licenses this role assignment under all its original conditions. Completeness says that every originally admissible occupant in this candidate fibre is trace-positive. The equation is proved from the original role and channel conditions; it is not obtained by defining those conditions to mean that the proposed trace is positive. No claim of exactness for other roles or unrelated inputs is required.

For example, if the original role is “the token at the independently identified receiver port of this occurrence,” the domain can specify a port-incidence relation and all joint compatibility conditions. A trace faithfully representing their conjunction is role-exact. Whether that relation has one incident token is a further fact, supplied by the architecture rather than the wording of the role. If the original role instead concerns a different port or an earlier occurrence, an identical positive bit is insufficient.

Lemma 4.3 (What the receiving bridge establishes). *Soundness of (4.1) licenses each positive trace assignment. Completeness excludes any originally admissible occupant outside the positive set. Thus role-exactness identifies that set with $F_\ell(d)$; declaration and cardinality alone do neither job.*

Proof. The first assertion is the forward implication and the second the reverse implication. Applying both pointwise gives equality of the two sets. An arbitrary singleton predicate may have its positive token outside $F_\ell(d)$, or omit another member of $F_\ell(d)$; neither possibility is excluded by its being a declared function. These are failures of the claimed receiver, not new sources of original authority. $\square$

Theorem 4.4 (Canonical qualification and exact channel realization). *Let W be the original family of candidate-input pairs $w = (d, s)$, with the independently interpreted J_ℓ of Definition 1.2. There is a unique Boolean qualification predicate*

$$\beta(d, s) = 1 \iff J_\ell(d, s).$$

For a specified joint channel encoding $\kappa : W \rightarrow K$, an exact reader $b : \kappa(W) \rightarrow \{0, 1\}$ with $b(\kappa(d, s)) = \beta(d, s)$ exists if and only if J_ℓ is constant on every κ -fibre. That reader is unique. It suffices that every original requirement A_u is constant on those fibres; the conjunction then supplies the qualification reader. Retention of original identity is a further sufficient case because every independently fixed original requirement is then recoverable.

Proof. Classical bivalence defines β from the already fixed truth of J_ℓ and forces the value of any other exact Boolean answer. If b exists, equal codes have equal answers, giving the fibre condition. Conversely, define

$$b(k) = 1 \iff \exists(d, s) \in W [\kappa(d, s) = k \wedge J_\ell(d, s)].$$

An original positive instance supplies a witness. Any witness at the same code transfers its qualification by fibre constancy, so the converse holds. Every code in the stated domain has a preimage; this also proves uniqueness without choosing representatives. Constancy of all A_u gives constancy of their conjunction. If the code retains original identity, equal codes retain the same original pair and therefore every such requirement. $\square$

This specializes Kernel Proposition 2.6 and Theorem 2.8, with same-question uniqueness as in Theorems 6.2–6.3 [2]. Thus complete qualification is derived from the frozen role; it is not an additional arbitrary axiom. The original semantic domain itself always realizes it, using the identity encoding. A proposed environmental implementation must still establish its interpretation and either the stated fibre condition with the constructed reader or the original incidence of its own answers. A different reader of the same code is not certified by the existence of an exact one. Nor does the semantic construction provide a physical channel, make J_ℓ decidable, or replace all original consumer content by one bit. A reader already proved to answer the same complete original qualification has discharged the bridge; no second completeness assumption is required.

Corollary 4.5 (Discharging the bridge from an original channel role). *Suppose the frozen role requires $C_d(s)$ as one of its original conditions, and the original architecture establishes every remaining requirement whenever $C_d(s)$ holds. If the trace faithfully reports that channel incidence, then it is role-exact.*

Proof. An originally admissible assignment satisfies every original requirement, including $C_d(s)$. Conversely, channel incidence and the architecture’s remaining-condition guarantees imply every $A_u(d, s)$, hence $J_\ell(d, s)$. Faithful channel reporting now gives (4.1) in both directions. $\square$

The channel’s necessity here comes from the original role, not from the proposed selector. In the baseline, the displayed trace definition already reports stable channel coupling exactly, while Adm is a separate field of the domain. Where the original role and its upstream proof identify those questions, the corollary discharges reception. Where they do not, a zero coupling bit is not yet a negative role-qualification answer. The checklist of declaration, stability, bivalence, and outcome independence alone supplies no such identification. This is the particular receiving issue, distinct from the corpus’s proved existence of canonical complete qualification.

Definition 4.6 (Lawful channel transformations and stability). The lawful transformations are identified from the complete original token, role, environment, and channel data before the proposed

trace is assessed. Their action preserves candidate membership and the fixed background assignments. Establish transport of the channel conditions and original requirements, hence

$$C_{gd}(gs) \iff C_d(s), \quad J_\ell(gd, gs) \iff J_\ell(d, s).$$

Where the trace is role-exact on the compared presentations, its stability is the consequent covariance law

$$\text{Tr}_{gd}(gs) = \text{Tr}_d(s). \quad (4.2)$$

At a fixed complete channel input d , its automorphism group H_d is the group preserving those original data. It is a subgroup of the automorphisms of the internal reduct. It is not defined merely as the permutations that happen to preserve a preferred trace output.

The implication from exact reception and incidence transport to (4.2) follows by applying (4.1) at d and gd . This receives AMetric Theorem 5.2's direction from original incidence to lawful representation [3]. Stability under a self-chosen group would not prove that the group contains all the lawful comparisons required by the original task.

Definition 4.7 (Outcome-independent authority). The channel source, role interpretation, and receiving argument have independent original-domain grounds. They do not obtain their first license from the preferred downstream confirmation, posterior, loss, utility, or success whose evidential role they are supposed to fix. Additionally, over the original family V of licensed variations of downstream evaluation, forgetting only those evaluative coordinates must retain both the trace and original role qualification:

$$r(v) = r(v') \implies (\text{Tr}_v(s) = \text{Tr}_{v'}(s) \text{ and } J_\ell(v, s) \iff J_\ell(v', s))$$

for corresponding candidates, with the complete anchor data and original conditions retained by r . The allowed variations and their meanings are fixed independently of the desired answer. This invariance requirement supplements the independent warrant requirement; it does not replace it.

If evaluation scores are determined by the retained data, the fibres of r may be singleton, making the displayed test vacuous. The independent warrant condition still excludes using the desired score as the first authority for the role. Raw physical outcomes, predeclared protocol conditions, and independently licensed numerical thresholds remain available when the original role requires them. Physical correlation with an outcome is not the prohibited dependence. The prohibited step is borrowing the role's missing license from the confirmational use that presupposes it.

Lemma 4.8 (No ungrounded mutual authorization). *In an independently sound rule system, let B be the grounded judgments and L a proposed family of role/channel licenses. If $B \cap L = \emptyset$ and every rule concluding in L requires a premise in L , then the least rule-closed set containing B contains no member of L .*

Proof. The complement of L contains B and is rule-closed: a rule with all premises in the complement cannot conclude in L . The intersection defining the least closure is therefore contained in that complement. This argument allows arbitrary premise sets. Independent grounding and sound interpretation are application premises, as in AMetric Theorem 7.4; a written license field does not prove them. $\square$

Definition 4.9 (Admissible environmental anchor). For the unique-original-occupant theorem, an admissible environmental anchor is a typed trace with an independently warranted joint channel source, the role-exact receiving proof (4.1), lawful transport (4.2), and outcome-independent authority. Its use retains the original candidate carrier, role, fixed other assignments, and complete consumers. It generates no new token and changes no candidate's original report or outcome interface. Singleton positivity is a separate condition, not part of this definition.

Remark 4.10 (Bivalence does not erase richer data). One specified qualification is bivalent. The joint channel may retain many numerical or nonnumerical properties. A threshold or ranking can enter a role predicate when its original meaning, warrant, and connection to that role are established. Merely declaring a threshold is insufficient; conversely, originally licensed richer data are not forbidden because they are quantitative. Only the resulting role-qualified bit is used by the singleton operator, and any other original consumers still retain their required data.

Lemma 4.11 (Admissibility of a received environmental anchor). *For an admissible environmental anchor, every trace-positive token yields an admissible R_s at this original role. No environmental oracle, independent token-generation step, or downstream confirmation license is used by that inference.*

Proof. The original joint incidence and soundness direction of (4.1) establish every requirement $A_u(d, s)$ for the assignment. Lawful transport retains those same requirements across presentations. Outcome-independent authority establishes their warrant without appeal to the downstream confirmation. The conclusion is an originally qualified role assignment, rather than qualification manufactured by a declaration. The remaining consumers are retained by the complete receiving specification. $\square$

Lemma 4.12 (A unique positive token is fixed by the input stabilizer). *If the trace is stable and $\text{Tr}_d^{-1}(1) = \{s^*\}$, every $h \in H_d$ fixes s^* .*

Proof. Covariance at $hd = d$ gives $\text{Tr}_d(hs^*) = \text{Tr}_d(s^*) = 1$. Candidate preservation and singletonity imply $hs^* = s^*$. $\square$

Lemma 4.13 (The trace does not select by downstream confirmation). *An anchor satisfying outcome-independent authority supplies its role warrant without using downstream confirmational preference. Allowed changes to that preference with the original anchor data fixed leave the role-qualified trace unchanged.*

Proof. The independent warrant requirement gives the first assertion; invariance on the original variation family gives the second. Neither assertion infers independence merely from correlation data or from the name of the channel. $\square$

5 The partial token operator and the class comparison

Definition 5.1 (Boundary-trace fixation operator). For the original singular token role, let $P_d = \{t \in \mathcal{S}_\ell : \text{Tr}_d(t) = 1\}$. Retain the partial operator

$$\text{BTFix}_{\text{Ch}}(\ell, \mathcal{S}_\ell; d) = s \iff P_d = \{s\}.$$

If P_d is empty or has several members, the operator is undefined. Its mathematical value records singleton positivity. An admissible selected assignment requires original-role soundness and the independent source and use license; Export S states that consequence. Claiming uniqueness among every originally admissible occupant additionally requires the complete receiving argument of Section 4.

Definition 5.2 (Environmental presentation and assignment). Write $X_{\text{env}}(\mathcal{S}_\ell; d) = \text{BTfix}_{\text{Ch}}(\ell, \mathcal{S}_\ell; d)$. When its qualified use returns s^* , set $R_{\text{env}} = R_{s^*}$. Outside the singleton case the operator returns no token, denoted $\uparrow$. This is semantic partiality of this operator, not a third truth value, a probabilistic preference, or a timeout in a proof search.

Lemma 5.3 (Fail-closedness of this trace rule). *When P_d is not singleton, returning a token as the value of this operator would depart from its graph. An alternative rule must disclose its additional criterion and warrant; it does not become this same trace rule by retaining its name.*

Proof. For empty P_d no positive token exists. For a non-singleton P_d , selecting one positive member requires discrimination not expressed by its membership in P_d alone. Both cases are outside Definition 5.1. A derived criterion may already be licensed by the complete original input. If so, it is assessed on that input; if it adds data or changes the original question, the change is explicit. Neither possibility totalizes the unchanged operator. $\square$

Proposition 5.4 (Stability of the environmental assignment). *A stable singleton trace fixes R_{env} at ℓ under the stabilizer of the complete input, including its background assignment. Across lawful presentations the assigned token transports covariantly.*

Proof. The fixed-input assertion is Lemma 4.12, together with preservation of the fixed other assignments. Equation (4.2) maps P_d bijectively to P_{gd} . If $P_d = \{s^*\}$, then $P_{gd} = \{gs^*\}$, giving the transported value and preserving the operator's domain. $\square$

Definition 5.5 (Original role equivalence). Let $i_\ell(s) = R_s$. For a receiver whose original question concerns role behavior up to equivalence, pull its independently justified assignment equivalence back to tokens:

$$s \equiv_\ell t \iff i_\ell(s) \sim_{\text{Role}} i_\ell(t), \quad q : \mathcal{S}_\ell \rightarrow Q = \mathcal{S}_\ell / \equiv_\ell.$$

The equivalence retains every original required consumer, including semantic availability and any continuation, transport, boundary, or failure question belonging to that receiver. Equality of Rep and Out alone is insufficient unless those interfaces cover the complete task. Lawful actions must preserve this equivalence before they are used on Q .

Theorem 5.6 (Exact trace descent and token lift). *A trace $b : \mathcal{S}_\ell \rightarrow \{0, 1\}$ descends to a unique $\bar{b} : Q \rightarrow \{0, 1\}$ if and only if b is constant on every q -fibre. When it descends,*

$$b^{-1}(1) = q^{-1}(\bar{b}^{-1}(1)).$$

If the class-positive set is $\{q^\}$, the same trace is raw-token singleton-positive exactly when $q^{-1}(q^*)$ is singleton. The quotient representation admits readers recovering every original consumer on the full original carrier precisely when those answers, including definedness, factor through q . For recovery only at one fixed class, constancy on that class's fibre suffices.*

Proof. The equation $b = \bar{b} \circ q$ implies fibre constancy. Conversely, set $\bar{b}([s]) = b(s)$; constancy makes this well-defined and surjectivity gives uniqueness. The inverse-image identity follows pointwise. In the singleton-class case the raw positive set is exactly the indicated fibre, proving the cardinality assertion. For each original totalized consumer a , a reader from Q exists exactly when $a(s) = a(t)$ whenever $q(s) = q(t)$, by the same factorization proof. Applying this argument to every original consumer proves the full-carrier assertion. Restricting the carrier to the chosen fibre proves the local version. Neither argument selects representatives. $\square$

If the original consumer asks which token identity occurs, that identity answer must also factor. A nontrivial fibre then cannot answer the original token question by itself. An arbitrary section of q may be useful as a calculational normalization, but does not recover which original token occupied a role. Further independently licensed lift data can determine that token; if already part of the original task, they must be retained, and if newly supplied, the enrichment must be stated. A raw singleton trace that splits an old equivalence class does not descend through that old quotient: its distinguishing work needs its own original license. Conversely, projecting an already fixed token to its class is legitimate once the receiver's identification is proved.

Remark 5.7 (Two types of fixation). The original operator answers a token question. A singleton role class can be a complete answer to a different original question without privileging a representative. Neither bivalence nor the word “fixation” identifies these result types. Several distinct tokens can each qualify for a use; this alone supplies no unique token occupant.

Export Q (Admissible fixation by a specified trace on the original role quotient). *For a class-valued receiver, let $q_d : \mathcal{S}_\ell \rightarrow Q_d$ be its original role quotient as in Definition 5.5, with the original qualification and every required consumer constant on its fibres. Let $P_d = \{s : \text{Tr}_d(s) = 1\}$. Suppose the trace has an independently warranted joint source and original-role use license, retains outcome-independent authority, and is sound:*

$$\text{Tr}_d(s) = 1 \implies J_\ell(d, s).$$

Suppose also that it is constant on every q_d -fibre and

$$q_d(P_d) = \{q_d^*\}.$$

Then the class-valued trace rule returns exactly q_d^ , its positive assignments are originally admissible, and the class supplies the required original consumer answers without choosing a representative. If the originally justified lawful action preserves the role equivalence and trace use and satisfies*

$$\text{Tr}_{gd}(gs) = \text{Tr}_d(s), \quad q_{gd}(gs) = gq_d(s),$$

the rule's qualified domain is invariant and its value transports as $q_{gd}^ = gq_d^*$. The stabilizer of the complete input fixes the class. This conclusion requires neither a raw singleton nor the reverse implication $J_\ell(d, s) \Rightarrow \text{Tr}_d(s) = 1$.*

Proof. Theorem 5.6 gives the descended predicate $\bar{\text{Tr}}_d$ and

$$P_d = q_d^{-1}(\bar{\text{Tr}}_d^{-1}(1)) = q_d^{-1}(q_d^*).$$

Thus the class operator has exactly one positive value. There is a positive representative, and soundness establishes its original qualification. Every original consumer has a well-defined reader

on Q_d by the same fibre argument, including its semantic domain; the reader evaluated at q_d^* is therefore the required answer for every representative in the positive fibre. This argument makes no representative selection. Covariance and the inverse lawful transformation give $P_{gd} = gP_d$, hence $q_{gd}(P_{gd}) = \{gq_d^*\}$. Preservation of the original trace-use license gives domain invariance, and fixing d gives class stability. The trace's authority is retained from its original source, not inferred from singleton cardinality. $\square$

The export proves uniqueness of the class fixed *by this trace*. The further assertion that every originally admissible assignment has that class requires $q_d(F_\ell(d)) \subseteq \{q_d^*\}$; soundness alone does not supply it. A receiver whose original target is a class need not recover a raw identity that it never requested. A receiver whose original target is a token still requires the lift in Theorem 5.6.

6 Boundary-Trace Fixation Theorem

Theorem 6.1 (Boundary-trace fixation of the original token role). *Fix $d, \ell, \mathcal{S}_\ell$, and the background assignment of Definition 1.1. In the internal non-selective setting, suppose:*

- (E1) *The original report/outcome interfaces are specified on their claimed domains, and the candidate outcome profiles are neutral.*
- (E2) *The original environmental input includes the identified environment and joint channel package.*
- (E3) *The trace is an admissible environmental anchor in Definition 4.9: its independent joint source, original-role soundness and completeness, lawful transport, and outcome-independent authority are established. The receiving bridge may be discharged by Theorem 4.4, Corollary 4.5, or another proof from the original role; it is not an extra assumption once that derivation is supplied.*
- (E4) *$P_d = \{s^*\}$ for some $s^* \in \mathcal{S}_\ell$.*

Then $X_{\text{env}}(\mathcal{S}_\ell; d) = s^$, R_{s^*} is admissible, and every admissible assignment R_t on this same candidate fibre and background has $t = s^*$. The assignment is stable at the original input and covariant across lawful presentations. It obtains no license from downstream confirmation. If, additionally, the complete joint internal-plus-indexical input is exhausted, boundary determination or boundary warrant is necessary for this fixation.*

Proof. By (E4), Definition 5.1 returns s^* . Soundness in (E3) gives $J_\ell(d, s^*)$, hence admissibility of R_{s^*} with all original joint requirements met. If R_t is any other originally admissible assignment in the same fibre, completeness gives $\text{Tr}_d(t) = 1$. By (E4), $t = s^*$. This proves existence and uniqueness of the original occupant, not merely uniqueness of the operator's displayed value.

Proposition 5.4 proves stability and covariance using the independently justified lawful action. Lemma 4.13 supplies the independence conclusion from the original grounds and allowed evaluation variations. The additional exhaustion hypothesis invokes Corollary 3.4 for necessity of the boundary source.

The elementary singleton implication does not require internal non-selectivity or outcome neutrality; those assumptions identify the problem for which environmental forcing is significant. Nor is indexical exhaustion inferred from neutrality. It is the separate joint hypothesis in the final sentence. $\square$

Proposition 6.2 (Sound partial certificates and the strength of uniqueness). *If only trace-to-role soundness is established, a singleton trace still gives one originally qualified assignment and a unique value of this trace rule. Uniqueness among every original admissible occupant additionally requires exclusion of all other members of $F_\ell(d)$; trace completeness is an exact way to discharge that obligation.*

Proof. Soundness places the returned token in $F_\ell(d)$. It does not constrain tokens outside the positive set. For instance, let $\mathcal{S}_\ell = \{a, b\}$, $F_\ell(d) = \{a, b\}$, and $P_d = \{a\}$. The trace is sound and singleton but two original assignments qualify. If completeness is established, every member of $F_\ell(d)$ lies in P_d , and singletonity excludes the second member. The example tests the inference; it is not an admitted counterexample to the theorem with (E3) established. $\square$

Export S (Sound licensed singleton selection). *Retain the original role, candidate carrier, background assignment and all required uses. Suppose the trace has an independently warranted joint source, its use retains outcome-independent authority, and positive soundness is established:*

$$\text{Tr}_d(s) = 1 \implies J_\ell(d, s).$$

If $P_d = \{s^\}$, the original trace operator returns exactly s^* and R_{s^*} is admissible. It is the only trace-positive assignment in this candidate fibre. If the original lawful action preserves that use license and the trace obeys $\text{Tr}_{gd}(gs) = \text{Tr}_d(s)$ on the claimed presentation family, the qualified operator domain is invariant, the value transports covariantly, and H_d fixes s^* . No completeness implication $J_\ell(d, s) \Rightarrow \text{Tr}_d(s) = 1$ is required for these conclusions.*

Proof. Singleton positivity gives the unique operator value by Definition 5.1; soundness gives $J_\ell(d, s^*)$ and hence the full originally required qualification of R_{s^*} . The covariance and inverse-transformation argument of Proposition 5.4 gives $P_{gd} = gP_d = \{gs^*\}$, while retention of the original source and use license preserves the qualified domain. At $gd = d$ the value is fixed. Outcome-independent authority provides the original warrant without downstream self-licensing. None of these steps asserts that a token outside P_d is inadmissible. $\square$

Here covariance must be justified from the original channel data and lawful transport. For example, faithful reporting $\text{Tr}_d(s) = 1 \iff C_d(s)$ and transport of C_d imply it directly, without requiring that C_d classify the entire role-qualification set. Merely choosing a group that preserves a preferred output would not justify the action. This sound export uses the source and positive receiving obligations, rather than the stronger role-exact anchor of Definition 4.9. Sufficient certificates need not equal the complete qualification relation [4, 8].

Export R (Agreement for an originally trace-required role). *Under Export S, suppose that the original role already requires the specified channel incidence, so that $J_\ell(d, s) \Rightarrow C_d(s)$, and the trace faithfully reports it: $\text{Tr}_d(s) = 1 \iff C_d(s)$. Then the trace is role-exact at this original input, and $F_\ell(d) = P_d = \{s^*\}$. Every licensed realization returning an occupant of this same original role and fixed background must return s^* . The receiving completeness follows from the original necessity; it is not a further independent assumption after that proof.*

Proof. Positive soundness gives $\text{Tr}_d(s) = 1 \Rightarrow J_\ell(d, s)$. Original necessity and faithful reporting give $J_\ell(d, s) \Rightarrow C_d(s) \Rightarrow \text{Tr}_d(s) = 1$. This is equation (4.1), as in Corollary 4.5. Set equality and singleton positivity yield the unique original occupant. Any licensed realization of that same role has its output in $F_\ell(d)$, so it agrees independently of its implementation. $\square$

Broader background compatibility may hold for several candidates while the original role also requires the trace. If that original role is $J_\ell(d, s) \iff K_d(s) \wedge C_d(s)$, it would be incorrect to require the trace to classify K_d alone. Conversely, adding C_d now to a previously fixed role that did not require it changes the task. A sound, independently licensed diagnostic singleton trace supplies Export S's conclusion, but cannot acquire the stronger agreement conclusion by being renamed a mandatory condition. The May baseline's universal uniqueness wording therefore retains a substantive same-role receiving obligation; it is not retrospectively reduced to uniqueness of the displayed rule value.

Corollary 6.3 (Exact fixation dichotomy for a received trace). *For a role-exact trace, $F_\ell(d) = P_d$. Consequently the original role has exactly one admissible occupant in this fibre if and only if BTFix_{Ch} is defined. Empty or multiple eligibility gives no unique occupant. Missing evidence about these conditions is an unresolved application obligation, not a proof that the original eligibility set is empty.*

Proof. Set equality follows from the two receiving implications. A set is singleton exactly when the operator's defining condition holds. $\square$

Corollary 6.4 (Residual automorphism collapse). *Under Theorem 6.1, H_d acts trivially on the fixed role occupant. Other candidates and other roles need not be fixed.*

Proof. Apply Lemma 4.12 to s^* . $\square$

Corollary 6.5 (Monotonicity under retained channel data). *When the environmental presentation retains the original internal data, its input stabilizer is a subgroup of the internal input stabilizer. If the qualified positive set is singleton, the token theorem applies under its remaining receiving hypotheses. A singleton positive orbit alone is not the singleton-set premise.*

Proof. Preserving the full input entails preserving its reduct, giving subgroup inclusion. The second assertion is the theorem. For the distinction, let a group fix a and exchange b, c in the positive set $\{a, b, c\}$. It has the singleton positive orbit $\{a\}$, but BTFix_{Ch} is undefined. Theorem 2.7 instead yields a unique equivariant selector if this is the full eligible family. That is a selector conclusion, not the assertion that a is the only originally eligible occupant. $\square$

Remark 6.6 (Practical anchoring). A detector identifier or calibration record can anchor a role when its warranted association to the original detector, occurrence, and record establishes the required incidence. An unattached identifier cannot. The theorem states this receiving condition; it neither infers the physical association from the identifier nor supplies the technology realizing it.

7 Converse, completeness, and lawful enrichment

Definition 7.1 (Environmental application obligations and failures). The following cover the points at which an alleged trace application must be assessed; several may fail together.

- C1. Original typing and authority.** Identify the original role, candidate carrier, other assignments, complete consumers, and independently warranted joint channel source; establish both directions of the receiving bridge when exhaustive occupant uniqueness is claimed.
- C2. Positive existence.** An empty trace-positive set gives no value of this operator.

- C3. Token uniqueness.** Several positives give no value of this singular trace operator. Test every already licensed joint criterion before claiming a global no-go.
- C4. Lawful transport.** Prove covariance under the original comparisons, including candidate membership and semantic domains.
- C5. Outcome-independent authority.** Ground the role and channel independently of downstream confirmational preference and preserve them under the licensed evaluation variations.
- C6. Parameter and auxiliary-data warrant.** Retain the original meaning and license of every threshold, ranking, marker, or auxiliary relation doing role work. A declared but unwarranted parameter is insufficient.
- C7. Original occurrence.** Distinguish a changed channel, role, or act from a new description or improved knowledge of unchanged original data.

Failure to discharge an obligation is not automatically a proved semantic failure. A false condition, a counterexample, and an absent proof have different evidential status.

Theorem 7.2 (Converse for complete environmental role fixation). *For the original singular role and fixed background, every complete warranted environmental realization of its unique occupant supplies a role-exact singleton relation on the original candidates. Conversely, any independently warranted boundary representation of that relation satisfying transport and outcome independence yields the token theorem. Thus, if no such licensed singleton relation exists anywhere in the complete original boundary input and its warranted consequences, no complete environmental realization of that unique-occupant task exists.*

Proof. A complete warranted realization of the unique-occupant task establishes that $F_\ell(d) = \{s^*\}$ from the original requirements. Its characteristic relation $b(s) = 1$ exactly when $J_\ell(d, s)$ is therefore singleton-positive. In an environmental realization its warrant is supplied through the accounted-for joint boundary relation. This is a normal form of an already warranted realization, not a procedure for obtaining that warrant by naming a desired singleton. Independent transport and outcome authority remain the stated receiving conditions. The reverse direction is Theorem 6.1. Negating existence of any such warranted representation gives the final implication. $\square$

This representation equivalence states what an exclusion proof must cover; it does not itself exclude every licensed boundary representation.

The quantifier over the entire licensed relation matters. Failure of one detector bit does not prove failure of every originally licensed boundary predicate. Deriving a conjunction or another consequence of original joint data may refine the trace without adding data. A criterion that merely chooses among several eligible occupants addresses a selection task; it does not prove that the other occupants were originally ineligible. The exact selector result is Theorem 2.7.

Corollary 7.3 (Symmetry obstruction on the complete input). *At any complete input z with $F(z) \cap \text{Fix}_S(G_z) = \emptyset$, no eligible equivariant partial selector whose domain contains z can return a token. A non-singleton trace alone does not establish this hypothesis.*

Proof. An equivariant selector's returned token must be fixed by G_z , by Lemma 2.6. The empty intersection excludes it. An equivariant endomap of the token set, such as its identity, is not a counterexample: it receives a token as input, which the unpointed fixation problem does not supply. $\square$

Theorem 7.4 (Closure by original semantic obligations). *For the fixed role and complete original input, distinguish the original eligibility relation $E(d, s) \iff s \in F_\ell(d)$ from the partial unique-occupant answer*

$$u(d) = s \iff F_\ell(d) = \{s\}.$$

An interpreted environmental proposal either satisfies all the obligations of its claimed result or fails at least one original semantic obligation. A complete realization is compared with the original graph of the question it actually claims: E for complete eligibility, or $G_u = \{(d, s) : F_\ell(d) = \{s\}\}$ for complete fixation. Defects include surplus or omitted records in that graph, failure of a claimed positive existence or singleton condition, loss of an original required answer on an encoded fibre, failure of transport or original-domain warrant, or substitution of a different question or occurrence. These are overlapping witnesses, not an exclusive finite taxonomy of physical mechanisms. If the necessary proof is missing rather than false, the claim remains unestablished.

Proof. For complete eligibility, let T_d contain the tokens whose assignments the proposal claims eligible. Exactness requires $T_d = F_\ell(d)$; failure of either inclusion supplies an omitted or surplus eligibility record. For complete fixation, instead compare the proposed graph K with G_u . Inequality supplies a record in one set difference. If complete outcome coverage is promised, compare with the totalized graph that reports a distinct unavailable marker precisely when $F_\ell(d)$ is not singleton. An empty or multiple eligibility set then correctly gives no token; that is not an omitted fixation answer. It does, however, refute a claim of a unique eligible occupant at that input. Exact representation of E alone proves no singletonity.

If the proposal claims to preserve a required answer through an encoding, the factorization proof of Lemma 3.2 gives exactly its fibre-constancy criterion, including availability. Failure supplies equal codes with different original answers. For each remaining original transport, compatibility, authority, and occurrence obligation, satisfaction or failure is evaluated under its independently specified meaning. A false universal claim has a failed instance; an invalid receiving inference must be corrected or independently justified. These checks cover the complete original obligations because their coverage was established before the proposal, not because a fixed number of failure names was listed. They instantiate the original-semantic exhaustion of [6]. $\square$

Corollary 7.5 (No retroactive environmental repair). *Fix an original occurrence and its complete role question. If it lacked the qualification needed for fixation, changing a channel, supplying a missing condition, or succeeding in a later act cannot make that unchanged original incidence qualified. Later evidence can establish an original qualification that was already present, or correct an earlier report about it.*

Proof. The original truth of $J_\ell(d, s)$ is fixed by the original conditions. If those data are unchanged, later description supplies no different value at that same incidence. If they change, the input or occurrence is different. A proof or observation of unchanged original data can improve knowledge without changing J_ℓ . Any separate requirement concerning evidence available at the earlier act retains its own historical data. This is the Kernel's historical non-substitution distinction [2], not a prohibition on new experiments or later certification. $\square$

Remark 7.6 (Richer inputs and new experiments). Lawful richer original inputs must be retained in the original theorem application. Missing them from an analyst's description does not make them newly created. Genuine new inputs, new protocols, or new acts may support new qualified constructions, all necessarily Kernel-governed. Their licenses do not flow backward into an unchanged earlier failure. A proof-search failure alone establishes neither original absence nor original multiplicity.

8 Class III and exact downstream reception

Definition 8.1 (Positive constraint and Class III). A positive role-anchor constraint is a warranted partial rule returning an originally eligible token, with its exact claimed domain and transport law retained. Class III environmental anchoring is the case in which boundary determination or boundary warrant is necessary after the complete non-boundary reduct is exhausted. For the complete unique-occupant result it is represented by a role-exact singleton boundary trace. A sound partial certificate is distinguished from this complete result.

Proposition 8.2 (Boundary information need not reduce to provenance). *Environmental anchoring is not universally reducible to a chosen provenance or indexical interface. If two licensed inputs have the same such interface and different original role answers due to their actual channel incidences, no function of that interface alone reproduces both answers. This does not prohibit jointly licensed dependence on provenance and channel data.*

Proof. The two inputs constitute a fibre collision for the proposed reduct. A function on the retained interface would have to give the same answer at both, contradicting the original answers. Conversely, joint dependence is permitted by Definition 3.1; no assumption of independent coordinate variation was made. $\square$

Proposition 8.3 (Local fixation). *Under Theorem 6.1, the original role ℓ has the unique qualified occupant s^* on its fixed background. The result makes no uniqueness claim about other roles or about all objects in the theory.*

Proof. The receiving proof and singletonity give $F_\ell(d) = \{s^*\}$. Other role fibres are not quantified over by this equality. $\square$

Continuation and Einstein role classes. Continuation DOI 10.5281/zenodo.20089824, Definitions 3.5–3.10 and Theorems 6.1 and 8.1, fixes one role-assignment class under complete role behavior, not a privileged representative [9]. Its displayed conclusion is uniqueness of the class fixed by the specified admissible trace. Einstein DOI 10.5281/zenodo.21616067, Definitions 18.4–18.7, Theorem 18.9 and Proposition 18.11, uses the corresponding class-level distinction for metric, source, vacuum, and horizon/certificate roles [10]. Its equivalence retains lawful continuation, redescription, boundary comparison, and failure handling. After identifying the role assignments via i_ℓ and retaining their original lawful action, these native equivalence definitions supply the required consumer constancy. Export Q gives the unique admissible trace-fixed class; neither raw-token singletonity nor completeness against every other admissible class is needed for that conclusion. A raw-token export instead requires the lift in Theorem 5.6. Einstein’s Theorem 30.1 separately assumes its boundary-fixed closed source ledger. The class export does not construct a physical channel for each named role or derive gravitational dynamics from the role ledger.

Continuation’s forcing premise also has its own receiving proof. Definition 3.6 excludes every admissibility-bearing fixing structure available in the original non-boundary role problem and expressly includes provenance, addresses and construction history. Corollary 4.6 assumes that strong class-nonselection condition. Let N retain all originally licensed joint non-boundary data and grounds. A warranted same-role class-fixing rule on N would itself be such an available structure, contradicting the definition, even if it uses a derived joint relation. Thus the complete joint exclusion required by Corollary 3.4 is already supplied by that original hypothesis; it is not an additional

assumption. The actual boundary relation is distinguished from these resources by Continuation’s Definitions 3.8–3.10, while its Definition 4.2 includes channel labels without identifying them with that relation. This argument gives necessity of boundary data or boundary warrant. It does not infer failure of bare value factorization, and it would not follow from weaker raw-representative nonselection or marginal tests.

Measurement’s singular record role. Measurement DOI 10.5281/zenodo.21306603 retains a singular branch-record assignment target. Its Definitions 3.2–3.5 and Theorem 3.6 type the occupant as a physically realized pair in $\text{Cand}(d) \subseteq B \times R$. Section 8, especially Definitions 8.2–8.3 and Theorem 8.5, requires a singleton sourced in the original measurement architecture and rejects an arbitrary singleton predicate [11]. Identifying the carrier and fixed assignment background, the trace’s admissibility-preserving use supplies the positive receiving obligation of Export S; its actual architectural license, occurrence, lawful transport, outcome independence and non-generativity remain required. Definition 3.4 already supplies six jointly required assignment conditions, including downstream use. Their interpretation and satisfaction must be warranted, but a new requirement family is not needed merely because those conditions are written in different notation.

The paper’s broader theory-level evidential criterion is agreement of the relevant ordering across original admissible assignments (Theorem 7.7); Proposition 7.10 permits multiple assignments in one standing-equivalence class. That endpoint does not always require one admissible raw token. A local sound singleton supplies a selected assignment; to infer the broader agreement, show that the original use family already follows that trace or prove the relevant consumer constant across all original admissible alternatives. Its quotient of assignment content does not silently replace the singular branch-record question. Where uniqueness of every original occupant is actually claimed, Export R or another complete receiving argument is required. The finite readiness constructor establishes its stated conditional trace result, not physical completeness from environmental correlation or decoherence.

Standing carrier identity. Standing Carrier DOI 10.5281/zenodo.22337281 fixes a target-complete continuation signature and its identity quotient [12]. Distinct signature classes remain distinct. Its Corollary 9.2 separately requires role eligibility, joint compatibility, lawful automorphisms, and a complete realization certificate. A singleton carrier class is consequently neither automatic environmental token fixation nor a replacement for its receiving conditions.

These comparisons preserve dependency direction. The present role-incidence, factorization, and selection proofs do not use downstream physical conclusions as premises. Class-valued and token-valued receivers can both be lawful when their original questions and connecting proofs are stated.

9 A structural witness and its limits

Example 9.1 (Two-token original-incidence model). Let $\mathcal{S}_\ell = \{s_A, s_B\}$. The internal reduct has identical candidate reports and outcome profiles and admits their exchange. Its original environment contains an independently identified port p with incidence relation $C = \{(s_A, p)\}$. The original role is the token incident at p in this occurrence; its other required compatibility conditions are satisfied at that incidence. The original requirement is therefore $J_\ell(d, s) \iff (s, p) \in C$. Set $\text{Tr}_d(s) = 1$ exactly when $(s, p) \in C$. The port and relation transform with a lawful presentation,

and the complete-input stabilizer must preserve them. The occurrence and port criterion obtain their interpretation independently of any downstream score.

Proposition 9.2 (The model occupies the singleton theorem). *The declared model has $P_d = F_\ell(d) = \{s_A\}$, and hence fixes R_{s_A} for its original role.*

Proof. Membership in the independently specified incidence relation gives both receiving directions. Its only incident token is s_A , establishing the singleton condition. Transport of the relation proves covariance, and its original definition has no downstream evaluative authority. The internal exchange ceases to preserve the complete input when the port relation is held fixed. Theorem 6.1 applies. $\square$

The witness is a set-theoretic model of the requirements. It does not prove that an arbitrary laboratory architecture realizes them. In particular, specifying a formal relation is legitimate for a mathematical model; identifying it with an actual physical channel requires independent evidence and domain reasoning.

```
def singleton_trace(candidates, trace):
    positive = [s for s in candidates if trace(s) == 1]
    return positive[0] if len(positive) == 1 else None

# A finite representation of the already specified model relation.
incidence = {("s_A", "p")}
tokens = ["s_A", "s_B"]
assert singleton_trace(tokens,
    lambda s: int((s, "p") in incidence)) == "s_A"
```

Remark 9.3 (What finite checks establish). The displayed computation checks the finite singleton rule. The general proofs above establish transport, fibre factorization, selector uniqueness, and the conditional role result. Neither code, metadata, a theorem label, a finite witness, nor a successful build establishes the independent interpretation of a physical channel or completeness of an arbitrary original consumer family.

10 Objections and the exact receiving burden

Proposition 10.1 (A warranted channel is not a naming convention). *Under the received singleton theorem, the token is fixed by its original role incidence, not by a freely chosen name. Formal stability alone is insufficient for this conclusion.*

Proof. The sound receiving direction establishes actual original qualification, completeness excludes every alternative qualified occupant, and transport preserves both assertions. Relabeling the represented token and channel together changes neither relation. A stable arbitrary marker lacking that bridge proves no corresponding role qualification. $\square$

Proposition 10.2 (Confirmation does not supply its own role). *Selecting the token because its downstream confirmation is preferred does not meet the independent-authority condition. A detection event or an independently warranted observation of the original channel is not excluded merely because it carries outcome information.*

Proof. The first rule uses the very evaluative conclusion that Definition 4.7 excludes as first authority. The second can establish the original incidence without that dependence. Each is assessed by its actual support and interpretation, not by whether its data are called outcomes. $\square$

Proposition 10.3 (Multiple positives do not answer the singular token question). *If $|P_d| \geq 2$, this trace has no value under Definition 5.1. Replacing its output by a class answers the original question only when the original consumers admit the factorization of Theorem 5.6. If the question requires raw token identity, a nontrivial class alone is insufficient.*

Proof. The first assertion is the operator definition. The factorization criterion proves the second. The identity consumer differs on distinct token members of a fibre, so cannot be recovered there. Several positive qualifications do not thereby become a graded qualification; they are qualifications of different tokens. $\square$

Proposition 10.4 (Calibration, discovery, and original occurrence). *Calibration that changes a failed original channel supplies a new or changed act. A later check proving that an unchanged original channel already met its conditions can establish that earlier qualification without creating it.*

Proof. The two cases differ in whether the original argument of J_ℓ changes. Corollary 7.5 applies pointwise. A historical requirement about available evidence remains separately attributable to the earlier occurrence. $\square$

Proposition 10.5 (Externality and component qualification are insufficient). *The environment's existence, or separate qualification of a token and an environmental component, does not establish their role-qualified channel relation. A received trace licenses only the uses warranted by that original joint relation.*

Proof. Separate component conditions do not imply an arbitrary joint condition: two separately eligible records may be incompatible for simultaneous use. The original family U_ℓ retains the joint requirement, and (4.1) must prove it for a positive token. Unrestricted facts about the environment have no automatic implication to that requirement. An additional licensed original relation may be relevant, but its connection must be proved. $\square$

Table 1: Existing closures with their exact receiving conditions.

Proposal	Required assessment
Choose the coupled-looking token	Establish original joint incidence and role reception. Appearance or a singleton inscription is insufficient.
Use the strongest coupling	Retain the original parameter's meaning and warrant; prove its role implication. An originally licensed numerical criterion is permitted.
Use the best-confirming token	It cannot obtain the role's first license from that downstream preference.
Combine two tied constraints	Assess their original joint license. Their conjunction can discriminate without introducing new data.
Use a singleton role class	Prove complete consumer descent; prove an independently licensed lift if token identity is required.

Proposal	Required assessment
Add later calibration	Changed conditions support a new act. Later discovery of unchanged qualifying conditions is improved knowledge.
Declare a preserving group	Derive the lawful comparisons from the original data; the chosen output cannot determine its own invariance test.
Invoke a closure theorem	Identify and prove its receiving hypotheses. A source name, registry, or finite computation does not discharge them.

11 Scope and challenge conditions

The main result is conditional on a role-exact, independently warranted channel. It proves one originally qualified token occupant on the fixed background. It does not prove the existence of a physical channel, select a physical outcome by dynamics, confirm a theory, or fix every role. The class comparison neither deletes the original token target nor prohibits calculational representatives. The environmental-forcing conclusion requires exhaustion of the complete joint non-boundary input, including all originally licensed derivations.

Bivalence concerns qualification at a complete use. It does not identify all positive objects, erase quantitative data, guarantee decidability, or turn a missing proof into a negative qualification. A semantically undefined singleton operator, an incomplete implementation, an unwarranted claim, and ignorance of an existing original relation must be distinguished. Lawful new constructions remain possible and necessarily Kernel-governed.

Remark 11.1 (Counterexample and application burden). A counterexample to a conditional mathematical implication must retain its stated hypotheses. A challenge to an application can instead show that the original role or candidate identification is wrong, a consumer is omitted, a joint warrant is absent, the lawful action is incorrectly chosen, a receiving implication fails, or a proof inference is invalid. Such a challenge need not manufacture a rival output satisfying the already strengthened theorem. Equally, disagreement with an output alone does not prove a scope change; a changed bearer, condition, question, or input must be exhibited.

12 Conclusion

A role-qualified singleton boundary trace fixes the original token role because its independently grounded positive set is exactly the original eligibility set. Singletonity supplies unique occupancy; lawful transport supplies stability; independent original authority prevents downstream confirmation from licensing its own role. The environmental route is forced only after the complete joint non-boundary input is exhausted.

A sound licensed singleton trace also supplies a valid selected assignment without excluding every other admissible alternative. On an original role quotient, singleton positivity fixes the class and its required consumer answers without selecting a representative under Export Q's original use and quotient conditions. These are distinct positive exports. When the original role already requires the specified channel, its necessity and positive license establish the stronger original-occupant result without another completeness postulate.

The exact selector, factorization, and class-lift results delimit this conclusion without suppressing

richer lawful inputs. A non-singleton primitive trace is not a universal selection impossibility; a singleton class is not automatically a token; a later proof of an original qualification is not its retroactive creation. Each distinction retains the same original bearer, complete question, and direction from incidence to faithful use.

A Source bindings and receiving ledger

Historical provenance. The authoritative baseline is the 25-page May 2, 2026 paper, version DOI 10.5281/zenodo.19988045. Its original bibliography and source map used unversioned corpus references and symbolic local labels. The cited authoring bundle **AASC System.zip** and the exact “Admissible Reference Fixation MAX Hardened V8” text have not been identified. Public pre-baseline editions of Reference Fixation (18531133), Structure (19657807), Kernel lineage (19844060), Exhaustion (19697219), Closure (19564661), and Measurement (19660161) provide separately pinned lineage evidence; they are not asserted to be the missing authoring bytes. The historical Standing–Admissibility, Bivalence, no-post-selection, no-deferred-admissibility, silent-redescription, and non-factorizability texts remain preserved in the preparation archive. Missing provenance is not by itself a mathematical counterexample.

Current receiving results. The following version bindings support the explicit scientific revision; none silently replaces a historical citation.

Table 2: Versioned source results and the local receiving work.

Source	Receiving condition and use
Kernel, 22968658	Proposition 2.4 and Theorem 3.1 supply necessary governance of non-degenerate determinate incidence. Theorem 2.8 and Theorems 6.2–6.3 require original interpretation, complete questions, and independently warranted answers. Lemma 3.3 and Section 9 distinguish original qualification, later knowledge, and a changed act.
AMetric, 22848051	Theorems 5.2 and 5.4 require original incidence transport, the actual input action, and eligible outputs. Theorems 7.2, 7.4, 7.5 and Section 7.6 separate fibre determination, grounding, and target-specific warrant. These are proved in the local forms used above.
Structure, 22874934	Sections 4–6 distinguish qualification from complete content, joint authorization, exact quotients, and safe interiors. The original role family here retains all required consumers; a positive bit does not replace it.
ATS, 22875011	Theorems 3.3–3.5 require complete original consumers for content factorization. Proposition 6.2 distinguishes a normalization section from recovery of an original representative. Theorem 6.3 retains joint applicability. The class bridge here proves only its stated descent/lift consequence.

Source	Receiving condition and use
Exhaustion, 22875070	Theorems 3.2–3.3 require original question coverage and exact readers; Theorem 4.8 tests joint constructors; Theorem 5.9 gives original semantic witnesses. Our closure quantifies over original obligations rather than claiming completeness from a finite registry.
Unus, 22875044	Theorem 3.1 and Corollary 3.3 distinguish one greatest safe region from positive existence; Sections 4–5 require faithful complete transport. Neither region uniqueness nor one qualification bit implies one token occupant. No interior theorem supplies the present singleton premise.
Closure, 22875348	Theorems 2.3, 3.3 and 4.2 distinguish graph exactness, information sufficiency, and joint-domain descent. Originally licensed richer inputs remain admissible subject to their own warrant. An incomplete implementation is not a semantic impossibility.

The historical repair and redescription closures already excluded retrospective creation of same-act standing, and the historical non-factorizability closure already required the joint relation. The revision exposes these conditions in the receiving proof rather than presenting the old oracle, post-selection, or component-only routes as newly discovered attacks. The complete historical withdrawal roster has not been recovered, so no claim about a new attack’s historical priority is made.

B Scientific scope of this revision

The original token carrier, role question, fixed background, and dependency direction are retained. Definition 5.1 keeps raw singleton positivity. The main scientific changes are explicit:

1. The unique-original-occupant conclusion exposes original-role soundness and completeness in (E3). Canonical complete qualification and its exact-realization criterion are derived in Theorem 4.4; Corollary 4.5 and Export R discharge the bridge from a matched original channel role. Identifying a particular channel predicate with that original qualification remains substantive relative to a literal reading of declaration, invariance, and bivalence alone. Where the original architecture already proves the match, no new completeness assumption is added. Proposition 6.2 and Export S preserve the admissible selected-realization conclusion for sound incomplete traces. The broader baseline uniqueness wording is not reinterpreted as merely one rule value.
2. Lawful transformations are supplied from the original data before stability is proved. Candidate preservation and the fixed background are explicit. Symmetry obstruction uses the actual input stabilizer, and a singleton set is distinguished from a singleton orbit.
3. Forcing tests the full joint internal-plus-indexical reduct. Source and failure classes are coverage, not an asserted disjoint partition. Derived original predicates remain available. A receiver whose original nonselection premise already excludes all such licensed structures, as in Continuation Definition 3.6, has supplied this joint condition rather than incurred an additional hypothesis.

4. Outcome independence concerns original authority as well as allowed evaluation variations. Original physical outcomes and warranted richer data are not erased.
5. Token/class descent and lift are proved for complete original consumers. Export Q preserves a native trace-fixed-class conclusion without imposing the stronger raw-token or all-admissible-occupant question. No class-only source is silently substituted for the singular role.
6. Later discovery is distinguished from later modification, and missing proof from proved failure. The original same-act repair prohibition is retained.

These changes remove inferences that were stronger than their displayed premises while retaining the strongest qualified token result and the exact more general selector result. No conditional theorem here discharges the empirical or mathematical receiving conditions of every concrete application.

References

- [1] Amos Jay Maley. *Boundary-Trace Fixation under Admissibility*. May 2, 2026 baseline. doi:10.5281/zenodo.19988045.
- [2] Amos Jay Maley. *Non-Degenerate Construction and the Kernel of Admissibility*. Reference edition September 26, 2026. doi:10.5281/zenodo.22968658.
- [3] Amos Jay Maley. *The AMetric Boundary as a Unique Regime-Invariant Constraint on Admissibility-Bearing Construction*. Reference edition September 20, 2026. doi:10.5281/zenodo.22848051.
- [4] Amos Jay Maley. *The Structure of Admissibility: Bivalent Qualification, Exhaustion, and the Unique Admissible Interior*. 2026. doi:10.5281/zenodo.22874934.
- [5] Amos Jay Maley. *Anchor, Tensor, and Skin: A Fixed-Domain Decomposition Theorem for Admissible Description*. 2026. doi:10.5281/zenodo.22875011.
- [6] Amos Jay Maley. *A Fixed-Domain Exhaustion Theorem*. 2026. doi:10.5281/zenodo.22875070.
- [7] Amos Jay Maley. *Unus Solus Possibilis Est: Corpus Closure and Uniqueness of the Admissible Interior*. 2026. doi:10.5281/zenodo.22875044.
- [8] Amos Jay Maley. *Closure by Exhaustion for Same-Scope Operators under Admissibility*. 2026. doi:10.5281/zenodo.22875348.
- [9] Amos Jay Maley. *Boundary-Trace Fixation of Continuation Loci under AASC*. May 2026. doi:10.5281/zenodo.20089824.
- [10] Amos Jay Maley. *Einstein Dynamics as the Unique Minimal First-Response Metric Projection of AASC Gravity*. 2026. doi:10.5281/zenodo.21616067.
- [11] Amos Jay Maley. *Admissible Record Construction in Physically Non-Selective Quantum Regimes*. July 11, 2026 edition. doi:10.5281/zenodo.21306603.
- [12] Amos Jay Maley. *Standing Carrier Identity Closure*. September 2026 edition. doi:10.5281/zenodo.22337281.